```
!pip install squidpy scanpy anndata pandas numpy matplotlib -q
```

```
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 79.5/79.5 kB 5.3 MB/s eta 0:00:00
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  Preparing metadata (setup.py) ... done
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    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.7/10.7 MB 93.1 MB/s eta 0:00:00
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.3/43.3 kB 2.7 MB/s eta 0:00:00
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    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 14.3/14.3 MB 76.1 MB/s eta 0:00:00
  Building wheel for docrep (setup.py) ... done
  Building wheel for pims (setup.py) ... done
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.3.3 which is incompatible.
```

```
import os
import matplotlib.pyplot as plt
import scanpy as sc

OUTPUT_DIR = "/content/outputs"
os.makedirs(OUTPUT_DIR, exist_ok=True)
_counter = [0]

def save_fig(name, fig=None):
    _counter[0] += 1
    path = f"{OUTPUT_DIR}/{_counter[0]:02d}_{name}.png"
    if fig is not None:
        fig.savefig(path, dpi=150, bbox_inches='tight')
    else:
        plt.savefig(path, dpi=150, bbox_inches='tight')
    print(f"✅ Saved: {os.path.basename(path)}")

# Prevent scanpy from auto-showing figures
# sq.pl.* does NOT support show= argument, so we use this global setting
sc.settings.autoshow = False

print("✅ Output folder ready:", OUTPUT_DIR)
```

```
✅ Output folder ready: /content/outputs
```

```
%matplotlib inline

import numpy as np
import pandas as pd
import anndata as ad
import scanpy as sc
import squidpy as sq

sc.logging.print_header()
print(f"squidpy=={sq.__version__}")
```

```
squidpy==1.8.1
```

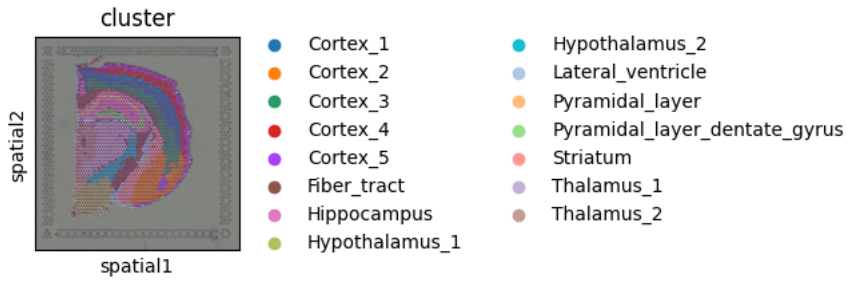
```
# Downloads pre-processed mouse brain Visium H&E dataset
img   = sq.datasets.visium_hne_image()
```

```
adata = sq.datasets.visium_hne_adata()

print("adata:", adata)
print("img:",   img)
```

```
INFO     Downloading visium_hne_image.tiff from
         https://exampledata.scverse.org/squidpy/figshare/visium_hne_image.tiff
Downloading data from 'https://exampledata.scverse.org/squidpy/figshare/visium_hne_image.tiff' to file '/content/data/images/vis
100%

 398M/398M

INFO     Downloading visium_hne_adata.h5ad from
         https://exampledata.scverse.org/squidpy/figshare/visium_hne_adata.h5ad
Downloading data from 'https://exampledata.scverse.org/squidpy/figshare/visium_hne_adata.h5ad' to file '/content/data/anndata/vi
100%

 329M/329M

adata: AnnData object with n_obs × n_vars = 2688 × 18078
    obs: 'in_tissue', 'array_row', 'array_col', 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts', 'log1p_total_cou
    var: 'gene_ids', 'feature_types', 'genome', 'mt', 'n_cells_by_counts', 'mean_counts', 'log1p_mean_counts', 'pct_dropout_by_c
    uns: 'cluster_colors', 'hvg', 'leiden', 'leiden_colors', 'neighbors', 'pca', 'rank_genes_groups', 'spatial', 'umap'
    obsm: 'X_pca', 'X_umap', 'spatial'
    varm: 'PCs'
    obsp: 'connectivities', 'distances'
img: ImageContainer[shape=(11757, 11291), layers=['image']]
```

```
sq.pl.spatial_scatter(adata, color="cluster")
save_fig("01_spatial_scatter_clusters")
plt.show()
```

```
✅ Saved: 01_01_spatial_scatter_clusters.png
```




```
# Scale=1.0 → immediate spot context
# Scale=2.0 → larger neighborhood context
for scale in [1.0, 2.0]:
    feature_name = f"features_summary_scale{scale}"
    print(f"Extracting {feature_name} ...")
    sq.im.calculate_image_features(
        adata,
        img.compute(),
        features="summary",
        key_added=feature_name,
        n_jobs=4,
        scale=scale,
    )
    print(f"  -> shape: {adata.obsm[feature_name].shape}")

# Combine both scales into one feature matrix
adata.obsm["features"] = pd.concat(
    [adata.obsm[f] for f in adata.obsm.keys() if "features_summary" in f],
    axis="columns",
)

# Remove any duplicate column names
adata.obsm["features"].columns = ad.utils.make_index_unique(
    adata.obsm["features"].columns
)

print("\nCombined features shape:", adata.obsm["features"].shape)
```

```
Extracting features_summary_scale1.0 ...
100%                                          2688/2688  [00:36<00:00,  186.17/s]
  -> shape: (2688, 15)
Extracting features_summary_scale2.0 ...
100%                                          2688/2688  [01:26<00:00,  35.99/s]
  -> shape: (2688, 15)

Combined features shape: (2688, 30)
```

```
!pip install squidpy scanpy anndata pandas numpy matplotlib leidenalg igraph -q
```

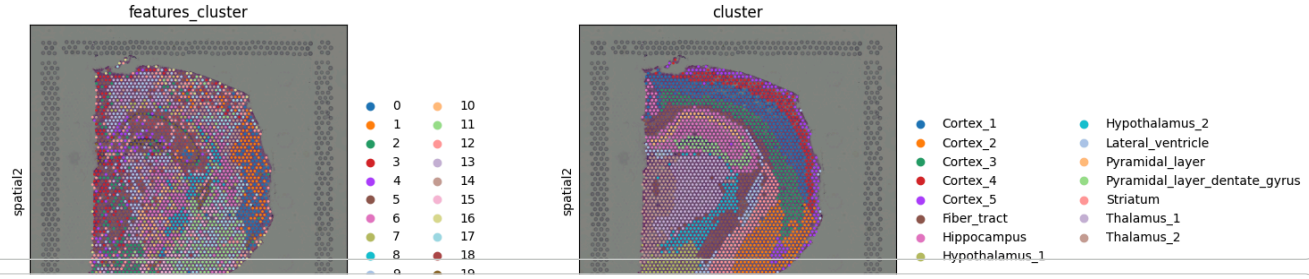
```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.7/2.7 MB 28.5 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.7/5.7 MB 42.9 MB/s eta 0:00:00
```

```
def cluster_features(features: pd.DataFrame, like=None) -> pd.Series:
    """Leiden clustering of image features. Filter columns with `like`."""
    if like is not None:
        features = features.filter(like=like)
    tmp = ad.AnnData(features)
    sc.pp.scale(tmp)
    sc.pp.pca(tmp, n_comps=min(10, features.shape[1] - 1))
    sc.pp.neighbors(tmp)
    sc.tl.leiden(tmp, flavor="igraph", n_iterations=2, directed=False)  # ← fixed
    return tmp.obs["leiden"]

# Cluster spots by image features
adata.obs["features_cluster"] = cluster_features(adata.obsm["features"], like="summary")

print("Feature clusters:", adata.obs["features_cluster"].unique().tolist())
```

```
Feature clusters: ['0', '3', '4', '5', '6', '1', '7', '9', '11', '13', '15', '12', '17', '10', '14', '18', '2', '8', '16', '19']
```

```
sq.pl.spatial_scatter(adata, color=["features_cluster", "cluster"])
save_fig("02_feature_clusters_vs_gene_clusters")
plt.show()
```

```
✅ Saved: 02_02_feature_clusters_vs_gene_clusters.png
```




```
sq.gr.spatial_neighbors(adata)
print("Spatial neighbors graph built.")
print("Connectivities shape:", adata.obsp["spatial_connectivities"].shape)
```

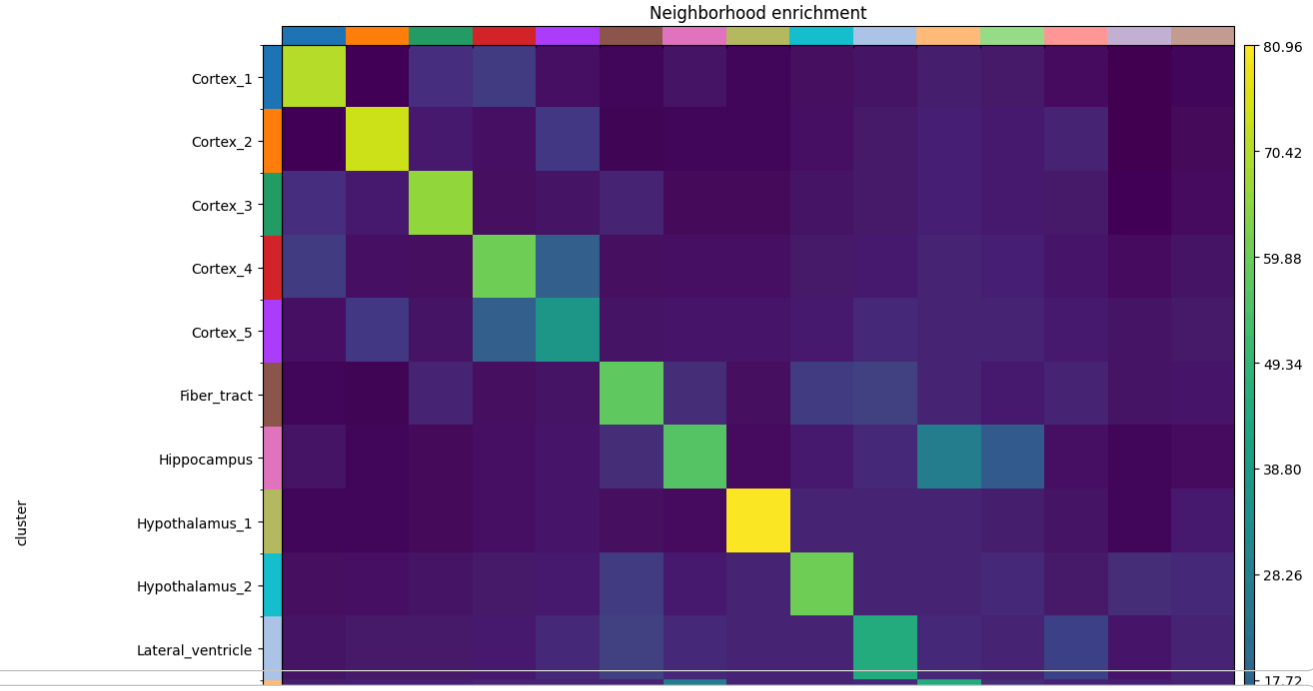
```
INFO     Creating graph using `grid` coordinates and `None` transform and `1` libraries.
Spatial neighbors graph built.
Connectivities shape: (2688, 2688)
```

```
sq.gr.nhood_enrichment(adata, cluster_key="cluster")
print("Neighborhood enrichment done.")
print("Results saved in adata.uns['cluster_nhood_enrichment']")
```

```
100%                                          1000/1000  [00:00<00:00,  1559.90/s]
Neighborhood enrichment done.
Results saved in adata.uns['cluster_nhood_enrichment']
```

```
sq.pl.nhood_enrichment(adata, cluster_key="cluster")
save_fig("03_neighborhood_enrichment")
plt.show()
```

```
Saved: 03_03_neighborhood_enrichment.png
```

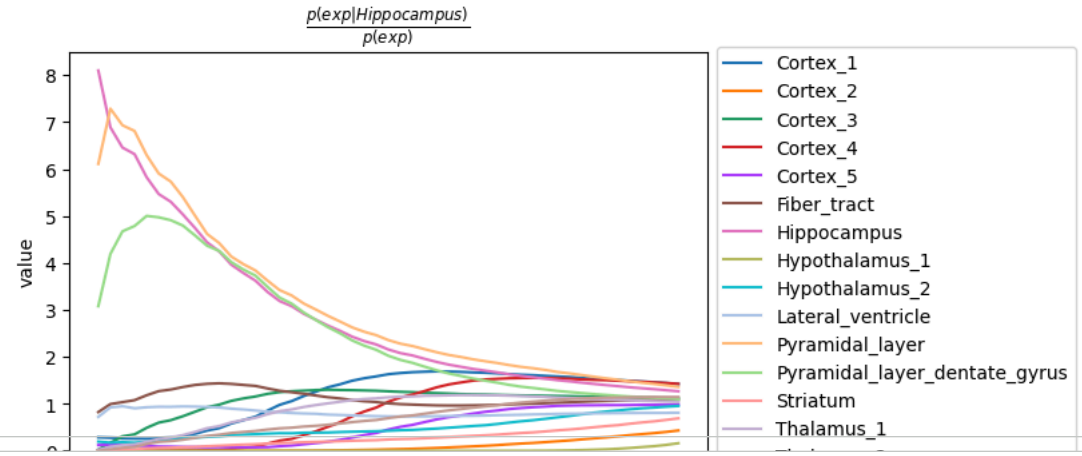



```
sq.gr.co_occurrence(adata, cluster_key="cluster")
print("Co-occurrence analysis done.")
```

```
Co-occurrence analysis done.
```

```
sq.pl.co_occurrence(
    adata,
    cluster_key="cluster",
    clusters="Hippocampus",
    figsize=(8, 4),
)
save_fig("04_co_occurrence_hippocampus")
plt.show()
```

```
Saved: 04_04_co_occurrence_hippocampus.png
```




```
sq.gr.ligrec(
    adata,
    n_perms=100,
    cluster_key="cluster",
)
print("Ligand-receptor analysis done.")
print("Results saved in adata.uns['cluster_ligrec']")
```

```
18.1M/? [00:00<00:00, 31.7MB/s]

1.78M/? [00:00<00:00, 4.63MB/s]

4.27M/? [00:00<00:00, 6.35MB/s]

100%                                          100/100 [00:35<00:00, 49.77permutation/s]

Ligand-receptor analysis done.
Results saved in adata.uns['cluster_ligrec']
```

```
sq.pl.ligrec(
    adata,
    cluster_key="cluster",
    source_groups="Hippocampus",
    target_groups=["Pyramidal_layer", "Pyramidal_layer_dentate_gyrus"],
    means_range=(3, np.inf),
    alpha=1e-4,
    swap_axes=True,
)
save_fig("05_ligrec_hippocampus_pyramidal")
plt.show()
```

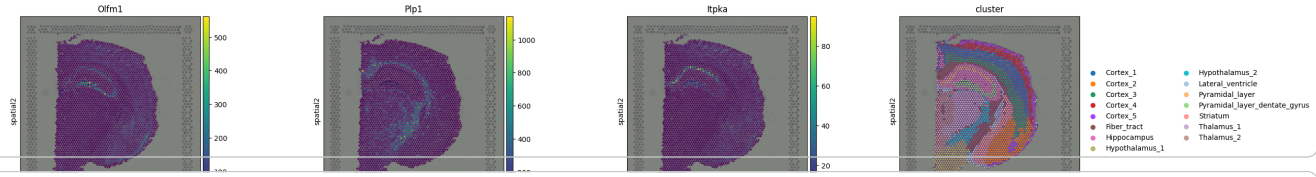
```
/usr/local/lib/python3.12/dist-packages/squidpy/pl/_ligrec.py:248: FutureWarning: The method uns_keys is deprecated and will be
  if cluster_key not in adata.uns_keys():
/usr/local/lib/python3.12/dist-packages/squidpy/pl/_ligrec.py:36: UserWarning: Over 500 categories found. Plot would be very lar
  super().__init__(*args, **kwargs)
✅ Saved: 05_05_ligrec_hippocampus_pyramidal.png
```

```
# Use top 1000 highly variable genes for speed
genes = adata[:, adata.var.highly_variable].var_names.values[:1000]

sq.gr.spatial_autocorr(
    adata,
    mode="moran",
    genes=genes,
    n_perms=100,
    n_jobs=1,
)

print("\nTop 10 spatially variable genes (Moran's I):")
print(adata.uns["moranI"].head(10))
```

```
100%                                          100/100 [00:43<00:00,  2.51/s]

Top 10 spatially variable genes (Moran's I):
               I  pval_norm  var_norm  pval_z_sim  pval_sim   var_sim  \
Olfm1   0.763291        0.0  0.000131         0.0  0.009901  0.000360
Plp1    0.747660        0.0  0.000131         0.0  0.009901  0.000306
Itpka   0.727076        0.0  0.000131         0.0  0.009901  0.000269
Snap25  0.720987        0.0  0.000131         0.0  0.009901  0.000255
Nnat    0.708637        0.0  0.000131         0.0  0.009901  0.000267
Ppp3ca  0.693320        0.0  0.000131         0.0  0.009901  0.000279
Chn1    0.684957        0.0  0.000131         0.0  0.009901  0.000290
Mal     0.679775        0.0  0.000131         0.0  0.009901  0.000198
Tmsb4x  0.676719        0.0  0.000131         0.0  0.009901  0.000218
Cldn11  0.674110        0.0  0.000131         0.0  0.009901  0.000304

        pval_norm_fdr_bh  pval_z_sim_fdr_bh  pval_sim_fdr_bh
Olfm1                0.0                0.0         0.011621
Plp1                 0.0                0.0         0.011621
Itpka                0.0                0.0         0.011621
Snap25               0.0                0.0         0.011621
Nnat                 0.0                0.0         0.011621
Ppp3ca               0.0                0.0         0.011621
Chn1                 0.0                0.0         0.011621
Mal                  0.0                0.0         0.011621
Tmsb4x               0.0                0.0         0.011621
Cldn11               0.0                0.0         0.011621
```

```
sq.pl.spatial_scatter(adata, color=["Olfm1", "Plp1", "Itpka", "cluster"])
save_fig("06_spatially_variable_genes_Olfm1_Plp1_Itpka")
plt.show()
```

```
✅ Saved: 06_06_spatially_variable_genes_Olfm1_Plp1_Itpka.png
```




```
import shutil
from google.colab import files

saved = sorted(os.listdir(OUTPUT_DIR))
print(f"\n📁 {len(saved)} plot(s) saved:")
for f in saved:
    print(f"   • {f}")

shutil.make_archive("/content/squidpy_visium_hne_plots", "zip", OUTPUT_DIR)
files.download("/content/squidpy_visium_hne_plots.zip")
print("✅ Download started!")
```

```
📁 6 plot(s) saved:
   • 01_01_spatial_scatter_clusters.png
   • 02_02_feature_clusters_vs_gene_clusters.png
   • 03_03_neighborhood_enrichment.png
   • 04_04_co_occurrence_hippocampus.png
   • 05_05_ligrec_hippocampus_pyramidal.png
   • 06_06_spatially_variable_genes_Olfm1_Plp1_Itpka.png
✅ Download started!
```